

CDS505

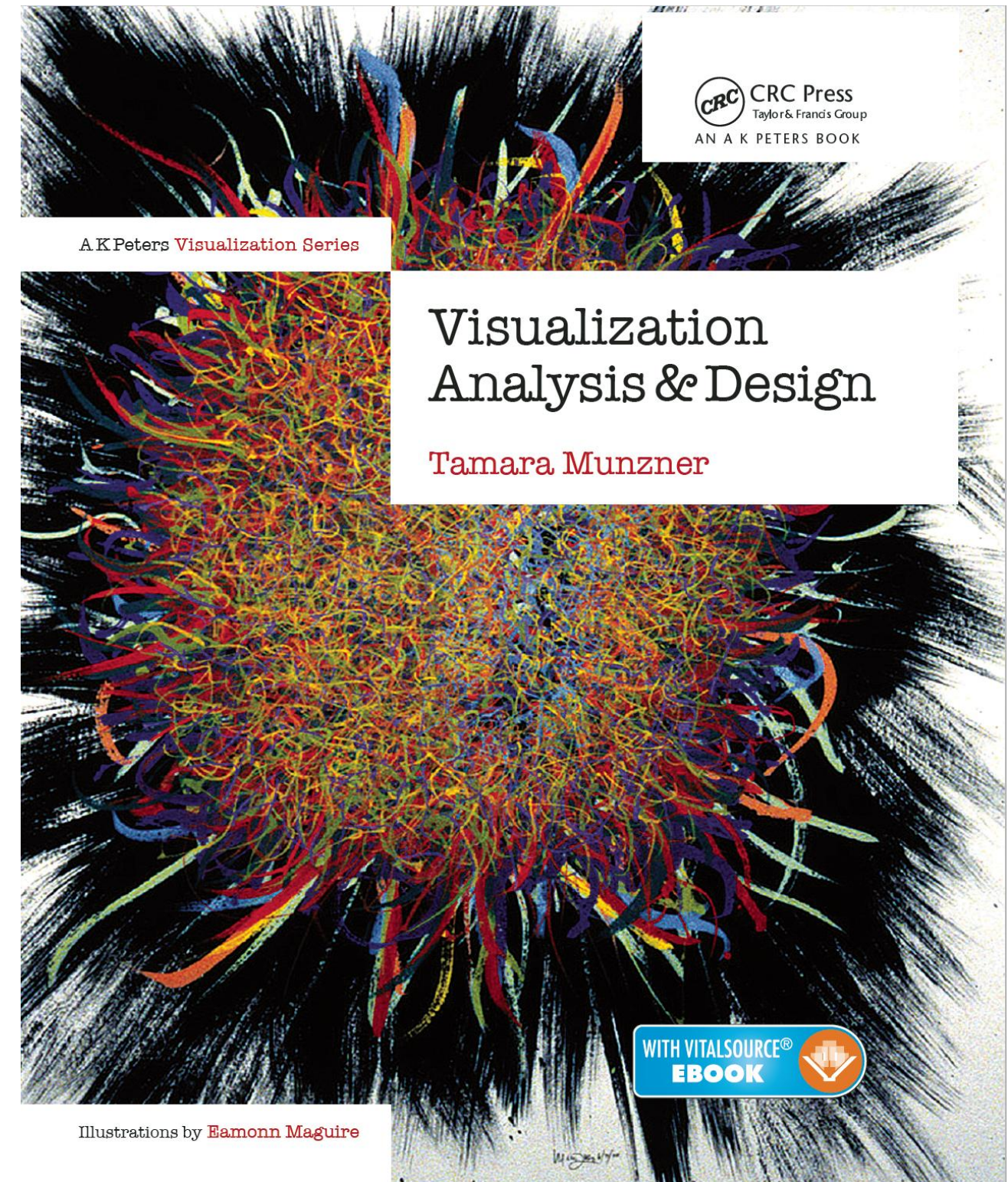
Data Visualization & Visual Analytics

Analysis & Validation

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Universiti Sains Malaysia*

Website:

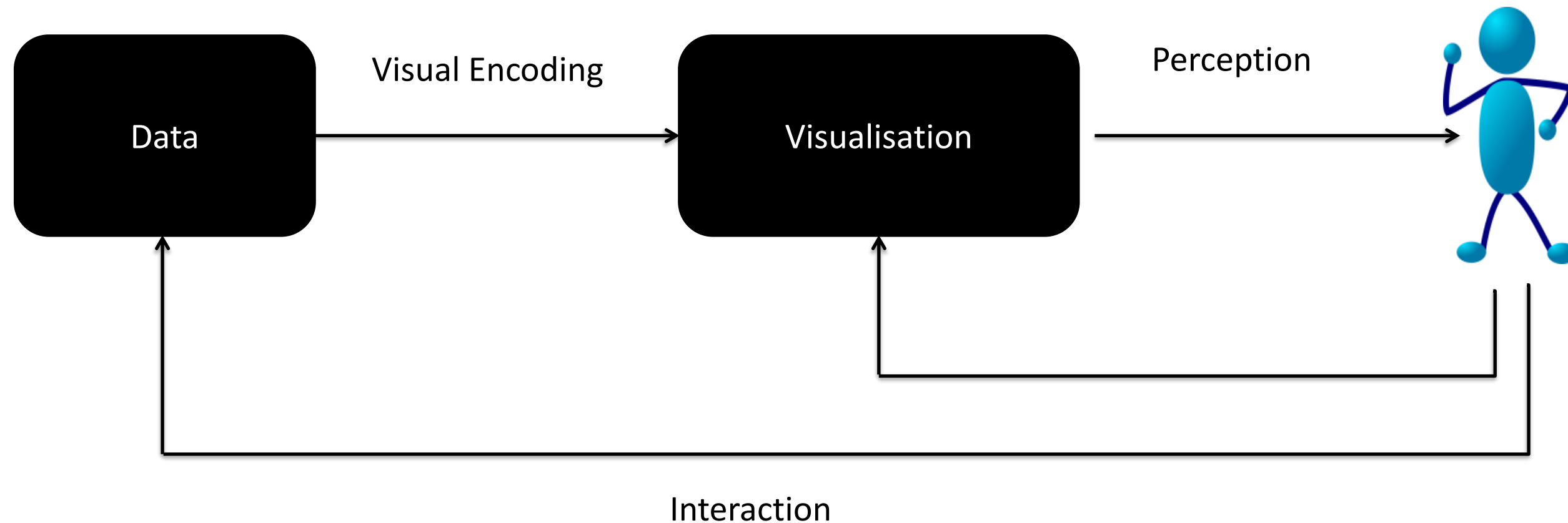
<https://cs.usm.my/index.php/faculty-member/194-nur-intan-raihana-ruhaiyem-dr>



- ✓ Recap previous class
- ✓ Analysis & Validation
- ✓ Class Activity

Visualization Definition: Revisit & Keywords

“Computer-based visualization systems provide **visual representations** of **datasets** designed to help people carry out **tasks** more **effectively**” (*Munzner*)

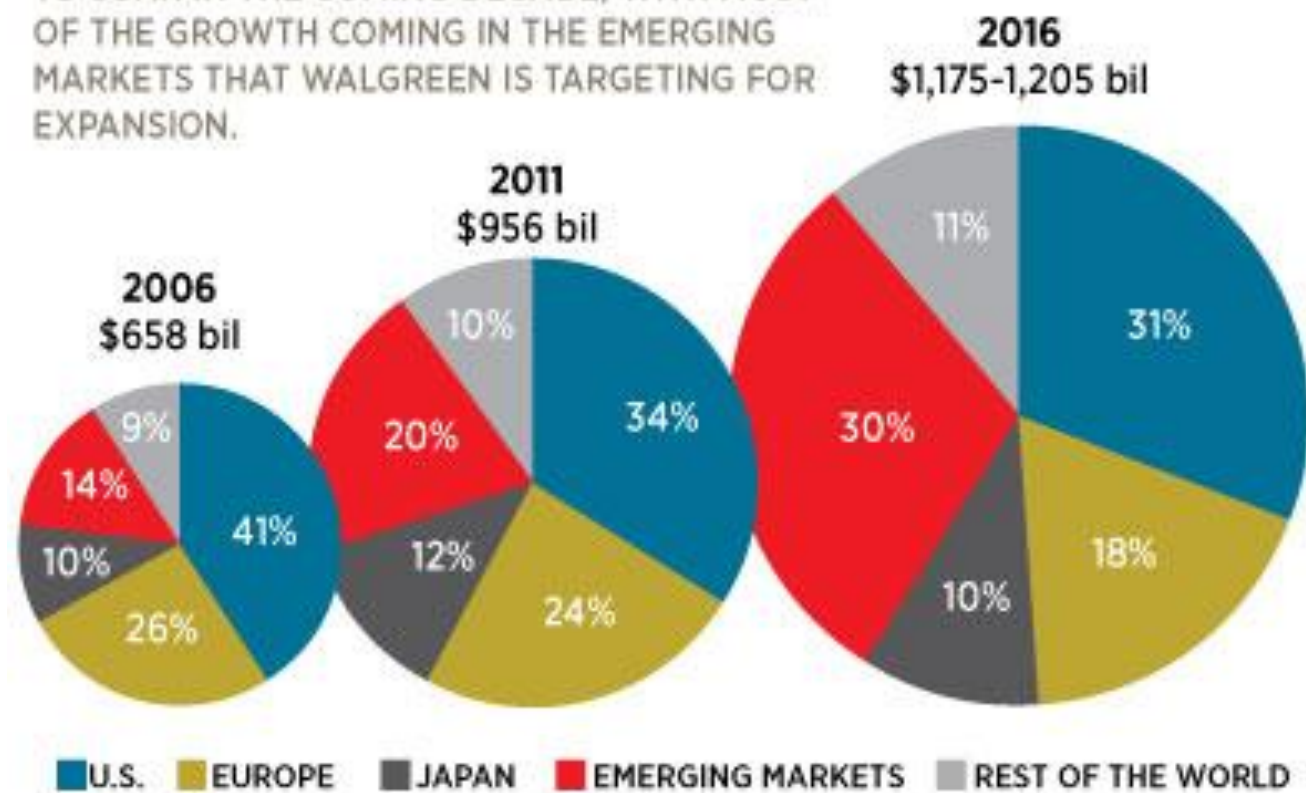


- How do we know how effective a visualization idiom is?
- Could we address this issue (measurement) in vis idiom's design?
- Why it is such as a challenging issue to address?
- What is a unit of measurement for “effectiveness” that we can agree on?

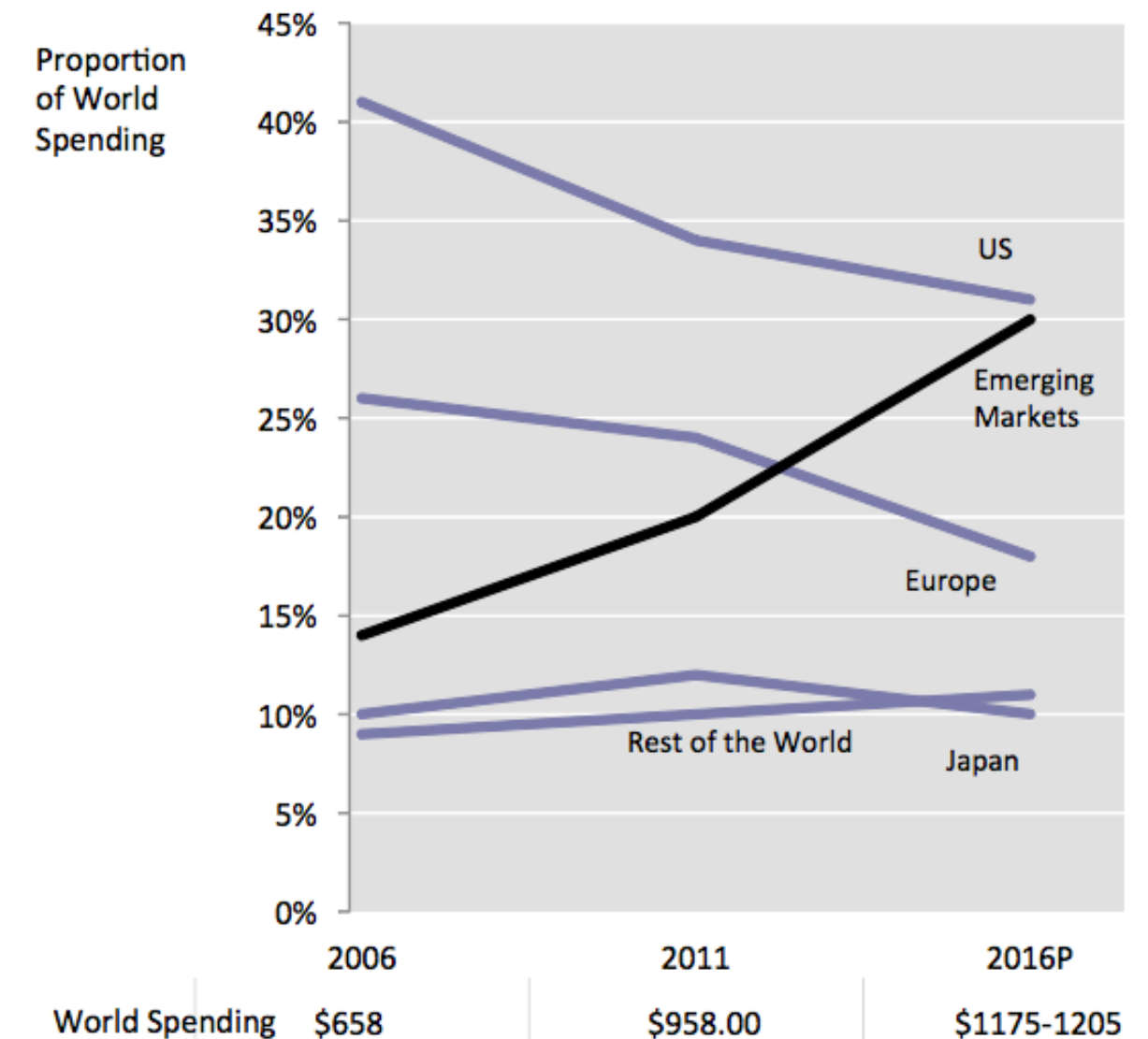
Expanding Circle of Errors

A WORLD OF DRUGS

GLOBAL SPENDING ON MEDICINE IS EXPECTED TO SOAR IN THE COMING DECADE, WITH MOST OF THE GROWTH COMING IN THE EMERGING MARKETS THAT WALGREEN IS TARGETING FOR EXPANSION.



SOURCE: IMS INSTITUTE FOR HEALTHCARE INFORMATICS.



Why validate?

 **Domain situation**
You misunderstood their needs

 **Data/task abstraction**
You're showing them the wrong thing

 **Visual encoding/interaction idiom**
The way you show it doesn't work

 **Algorithm**
Your code is too slow

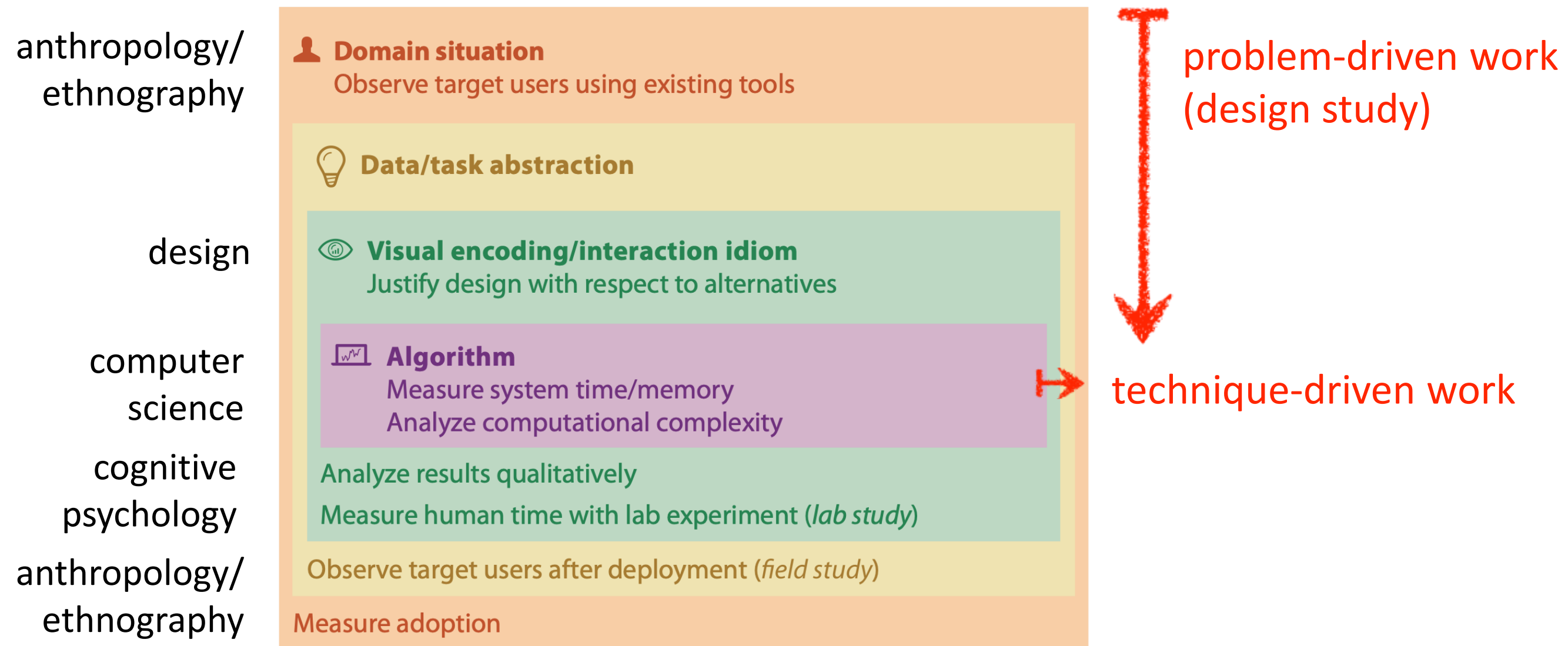
- Too many choices, most of them ineffective
- Validation refines design choices based on the feedback

Why it is difficult?

- Different ways to get it wrong at each level (user's requirements, wrong data & task abstraction, incorrect vis idiom, etc.)
- Require multi-disciplines validation approaches in different levels

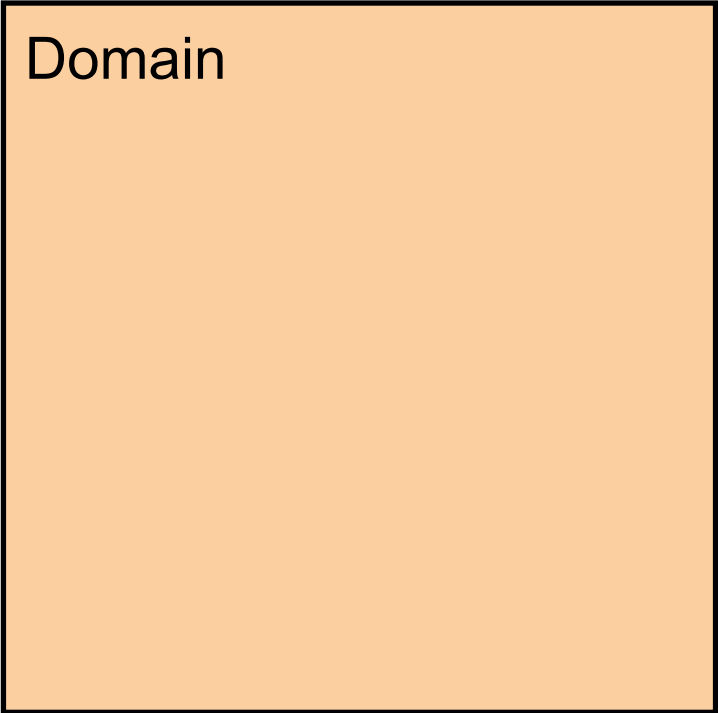
Why is validation difficult?

Solution: use methods from different fields at each level



Analysis Framework: Four level, three questions

- ***Domain situation***
 - who are the target users?



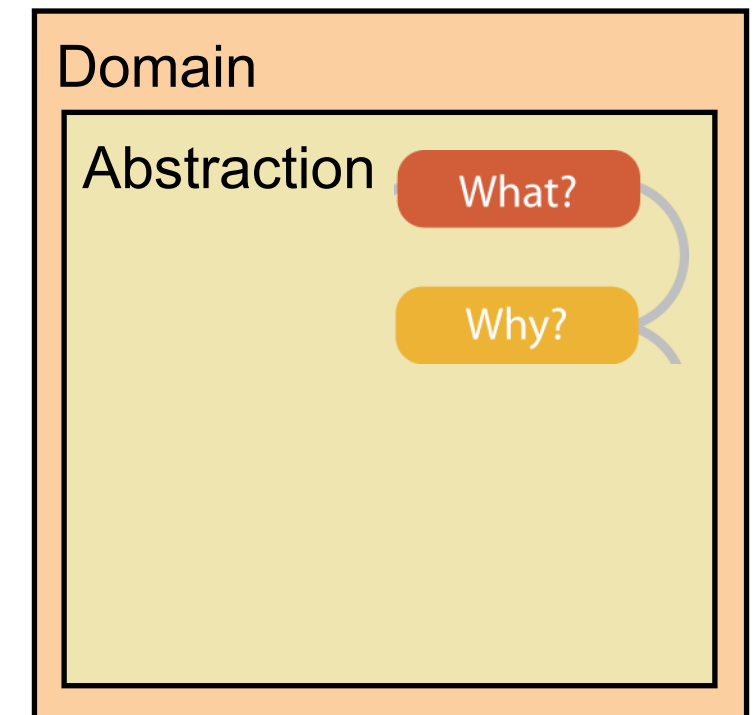
Domain

[A Multi-Level Typology of Abstract Visualization Tasks. Brehmer and Munzner. IEEE TVCG 19(12):2376-2385, 2013 (Proc. InfoVis 2013).]

[A Nested Model of Visualization Design and Validation. Munzner. IEEE TVCG 15(6):921-928, 2009 (Proc. InfoVis 2009).]

Analysis Framework: Four level, three questions

- **Domain situation**
 - who are the target users?
- **Abstraction**
 - translate from specifics of domain to vocabulary of vis
 - **what** is shown? **data** abstraction
 - **why** is the user looking at it? **task** abstraction

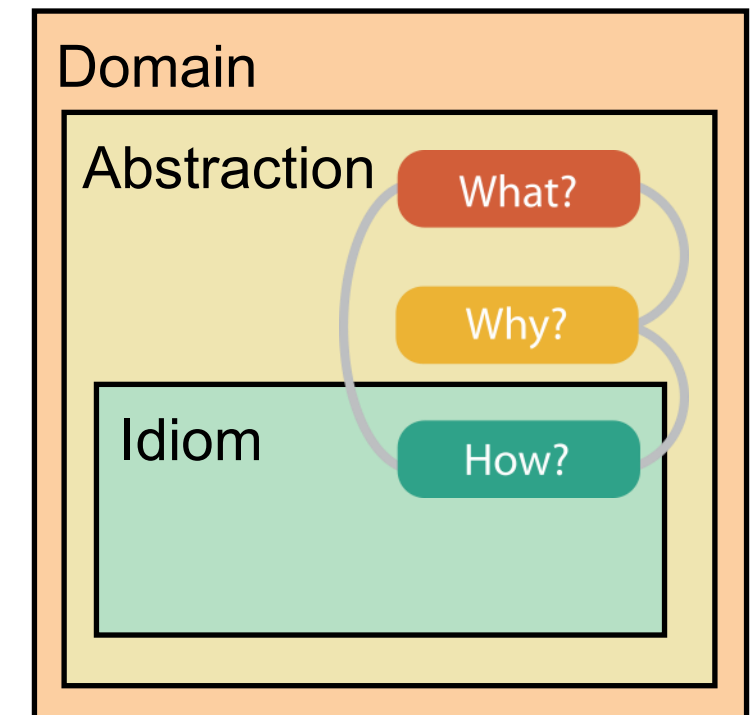


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 - **how** is it shown?
 - **visual encoding** idiom: how to draw
 - **interaction** idiom: how to manipulate

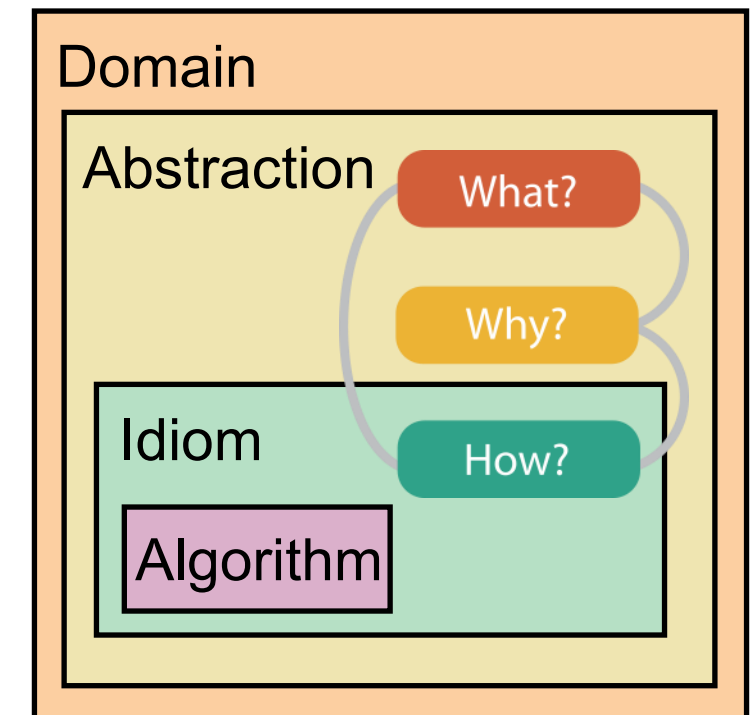


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- **Algorithm**
 - efficient computation



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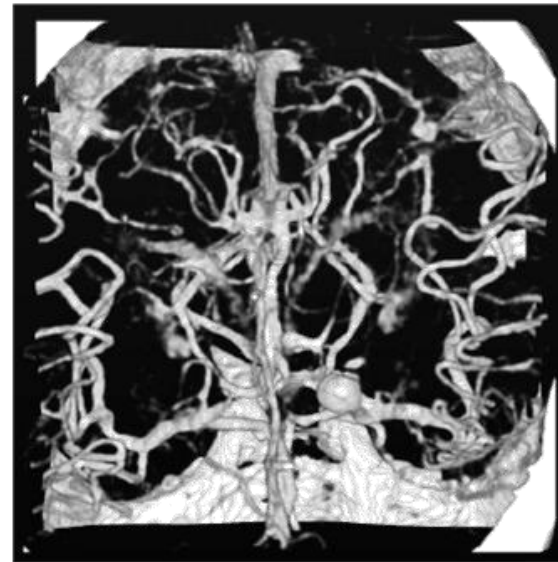
Comparison of Ray Casting and Texture Approaches

VTK



Ray casting

OGLE



Texture-based

<http://www.cora.nwra.com/Ogle/>

Amira



Texture-based

<http://www.amiravis.com>

VLIB



Ray casting

<http://vg.swan.ac.uk/vlib>

Close Up

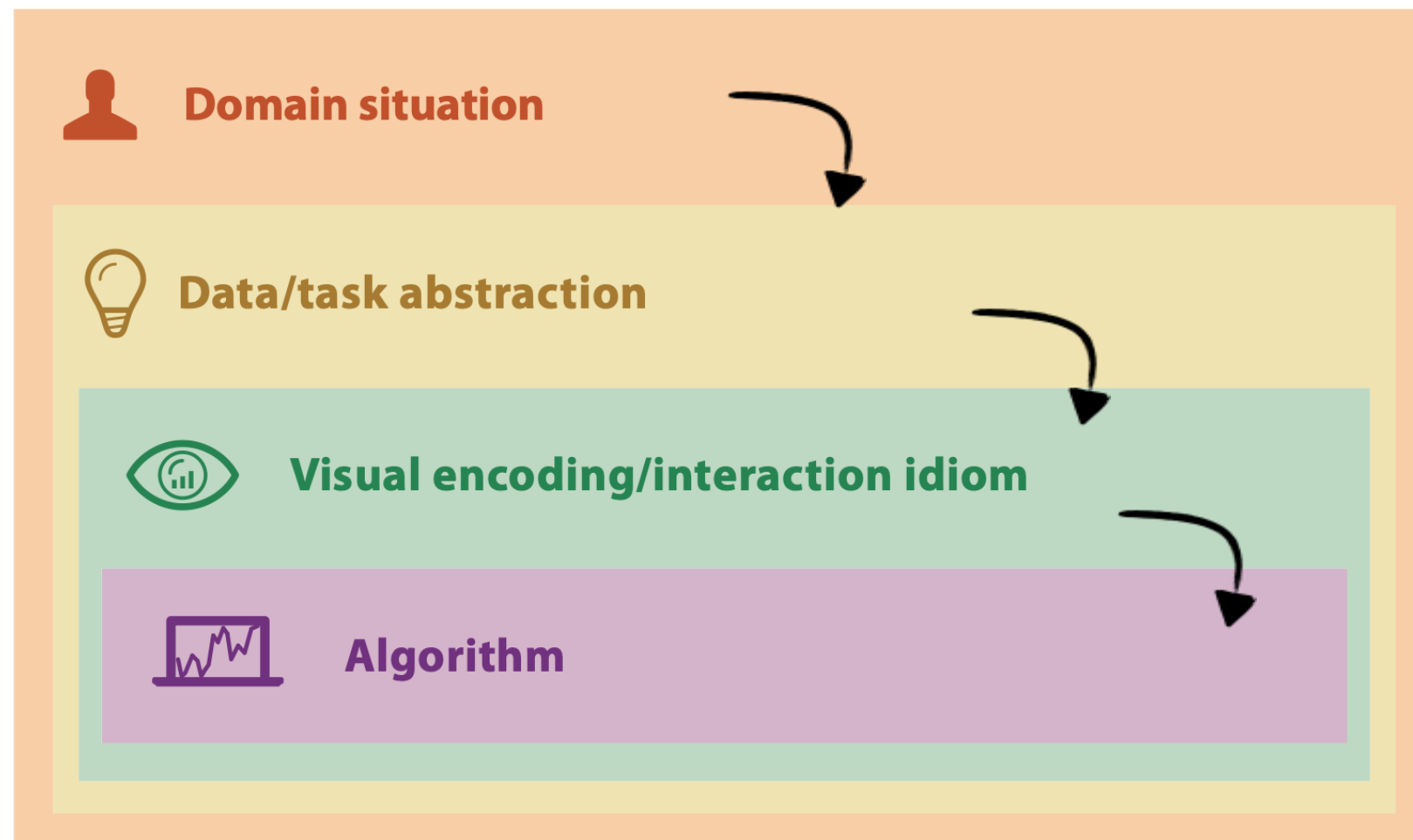


Ogle: texture-based



Vlib: ray casting

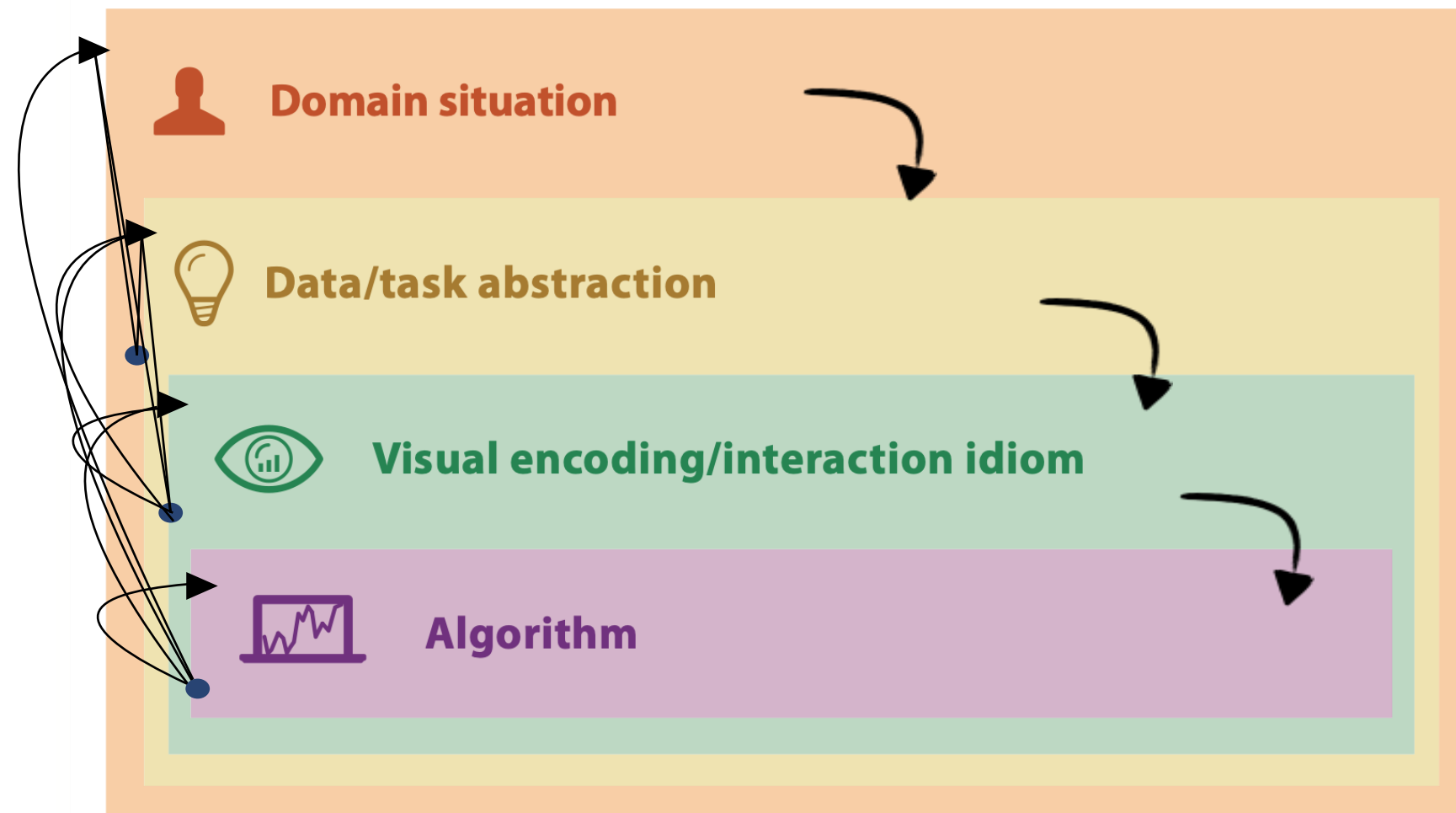
- Downstream: Cascading effects



- **Top-Down:** starts with user's goals and work through 4 design levels to create a solution that addresses user's goals effectively
 - Selecting existing vis idioms (cf designing new ones)
 - Challenges lies in abstraction level (mapping domain-specific to generic reps)

Nested Model

- Downstream: Cascading effects
- Upstream: Iterative refinement



- **Bottom-Up:** start with ideas on NEW vis idioms & algorithm; analyse existing idioms to improve computational efficiency (algorithm), coming up with new vis idiom design addressing specific gaps, etc.

Revisiting Validation – its Pitfalls/Threats

 **Domain situation**
You misunderstood their needs

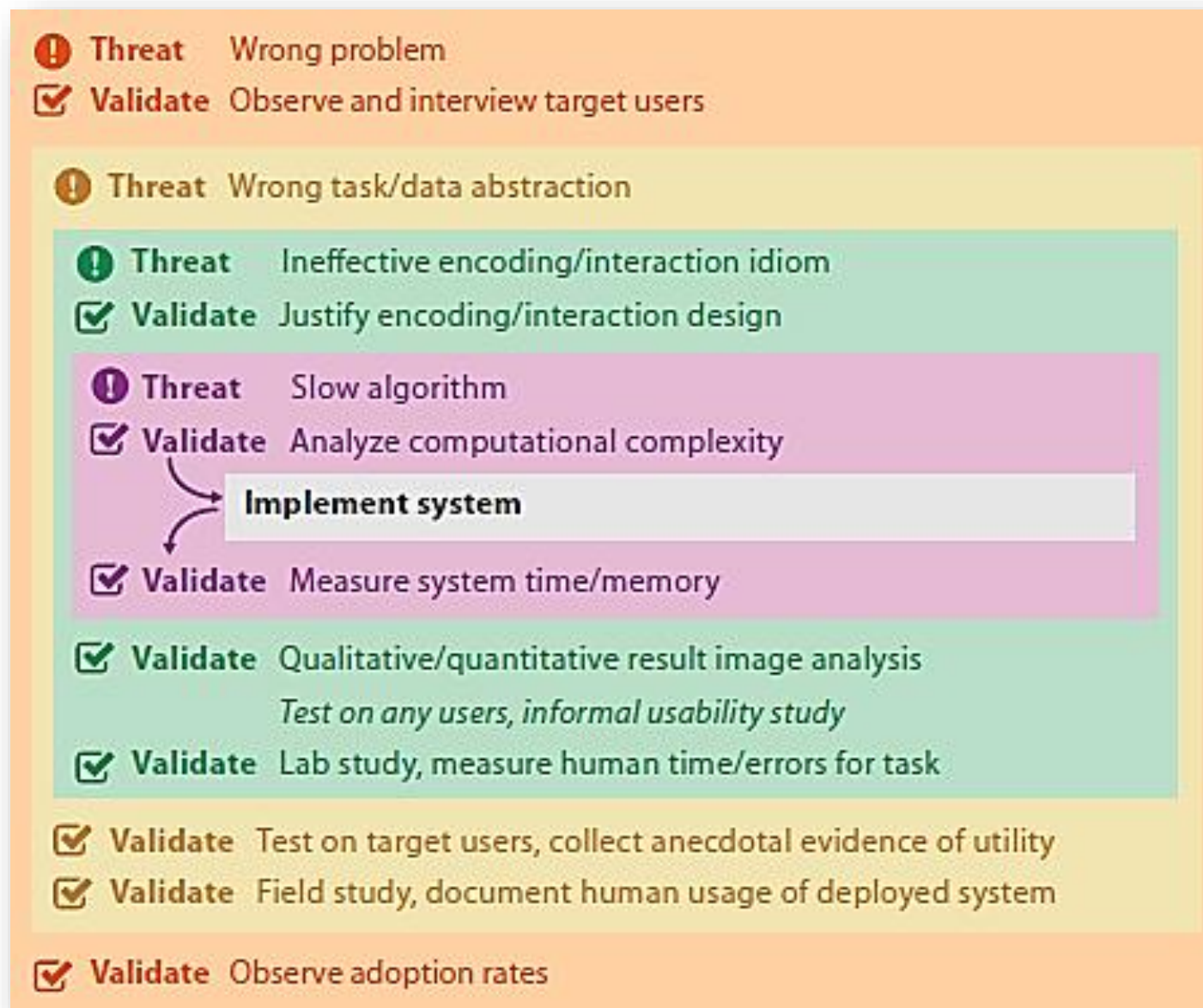
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- **Validation** – test if you've designed the
- right solution (at every design level)
- Checking your design decision using the same 4 levels model
- Distinct fundamental design decision errors at each level:
 - *Wrong Problem: Vis Designer misunderstood target's user needs*
 - *Wrong Abstraction: Vis Designer showing target's user the wrong thing.*
 - *Wrong Idiom: The way Vis Designer show the thing that does not work*
 - *Wrong Algorithm: Vis Designer code is too slow*

Validation Approaches - Overview



Immediate Approach :

Validation done at a level independent from other levels (no dependency)

Down-Stream Approach :

Validation at a level only possible when a complete instantiation of vis idiom is available i.e. need results from upper levels

Please explain the approaches of validation in each level:

- 1. Domain** validation approaches
- 2. Abstraction** validation approaches
- 3. Idiom** validation approaches
- 4. Algorithm** validation approaches

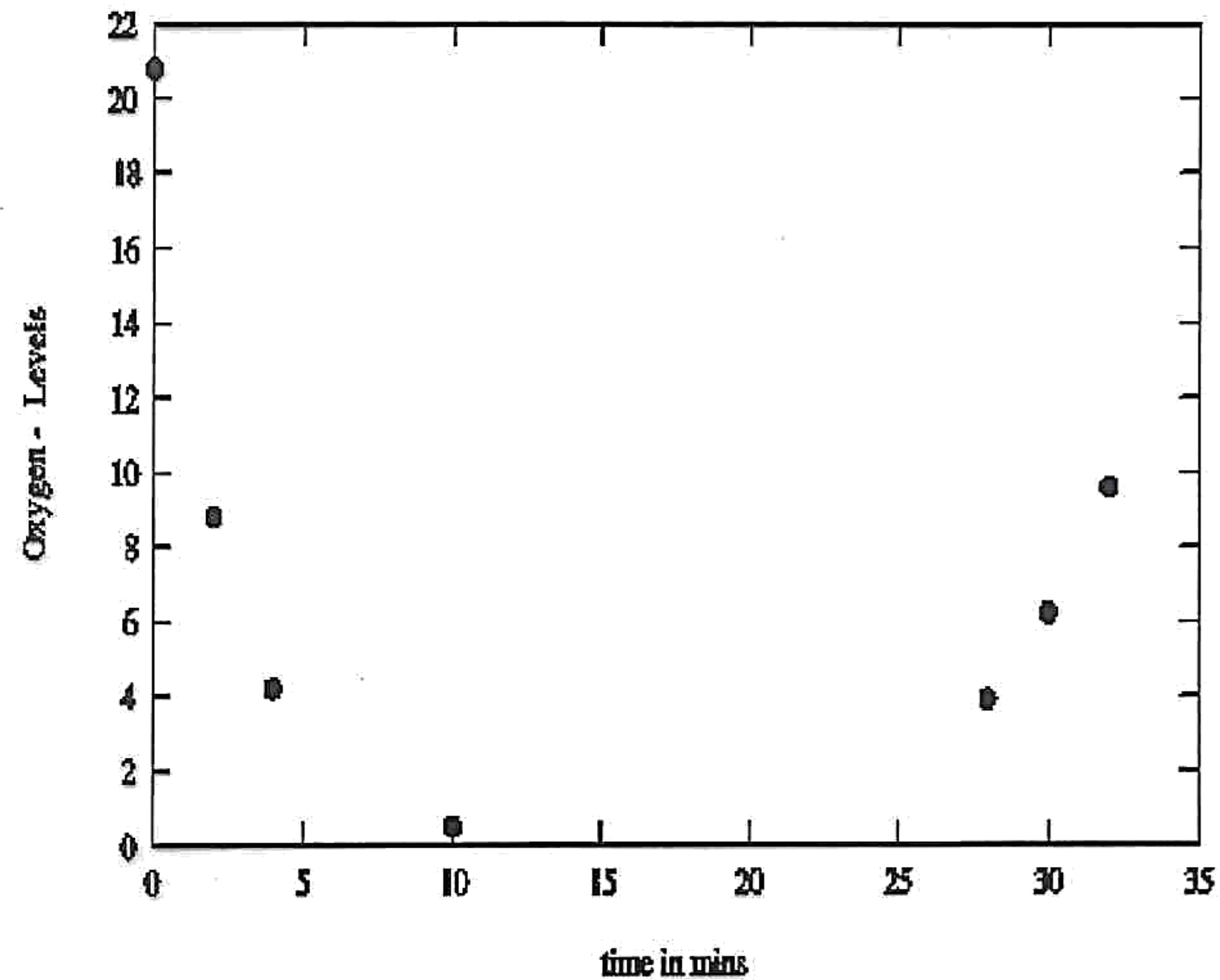
A Simple Example – Algorithm Incorrectness

This table shows the observed oxygen levels in the flue gas, when coal undergoes combustion in a furnace.

TIME (mins)	0	2	4	10	28	30	32
OXYGEN (%)	20.8	8.8	4.2	0.5	3.9	6.2	9.6

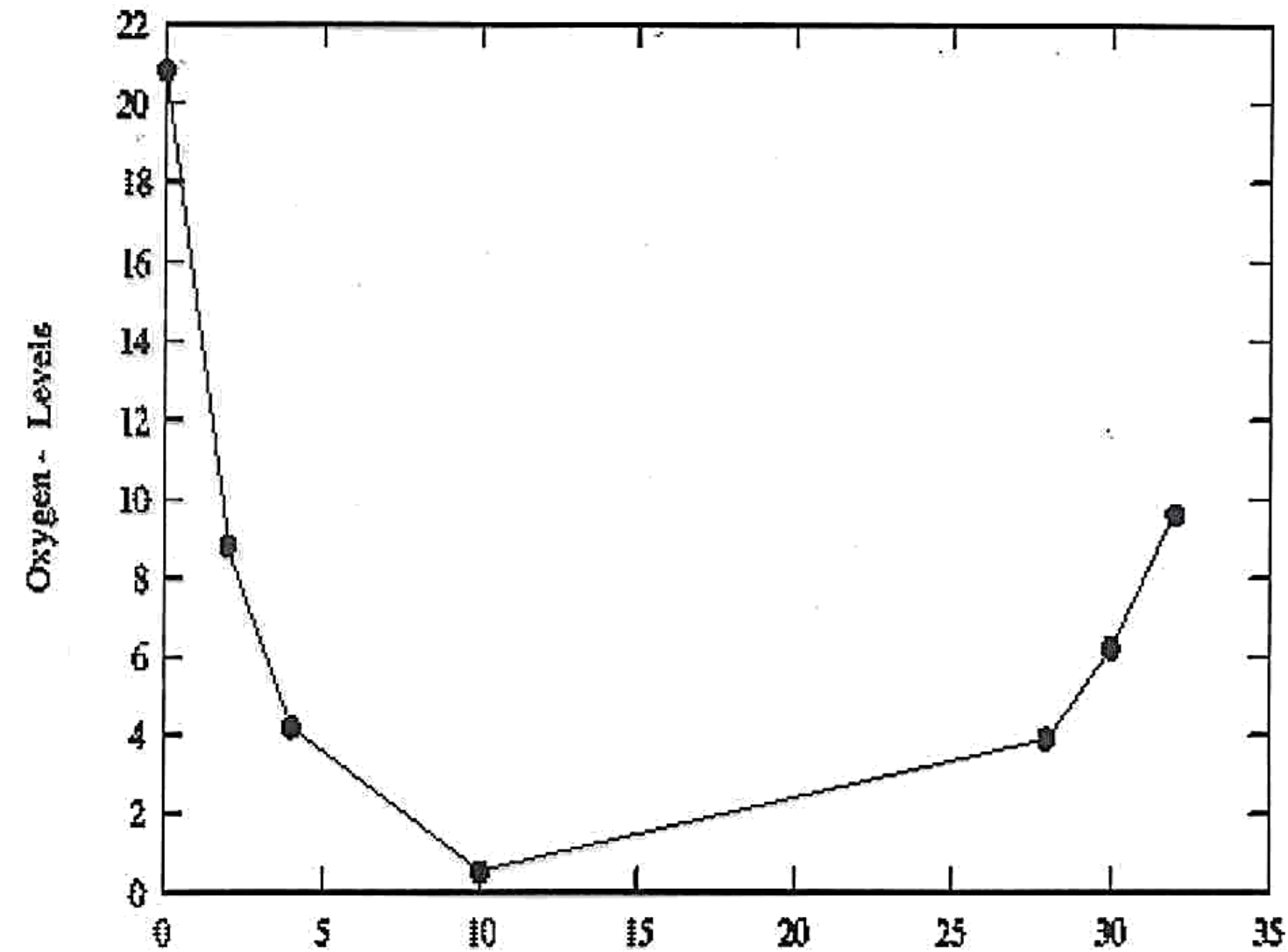
User would like to investigate if there is any correlation between these two variables; what kind of correlation?. Can you design a believable vis idiom to address user's needs?

Visualizing the Data – Scattered Plot



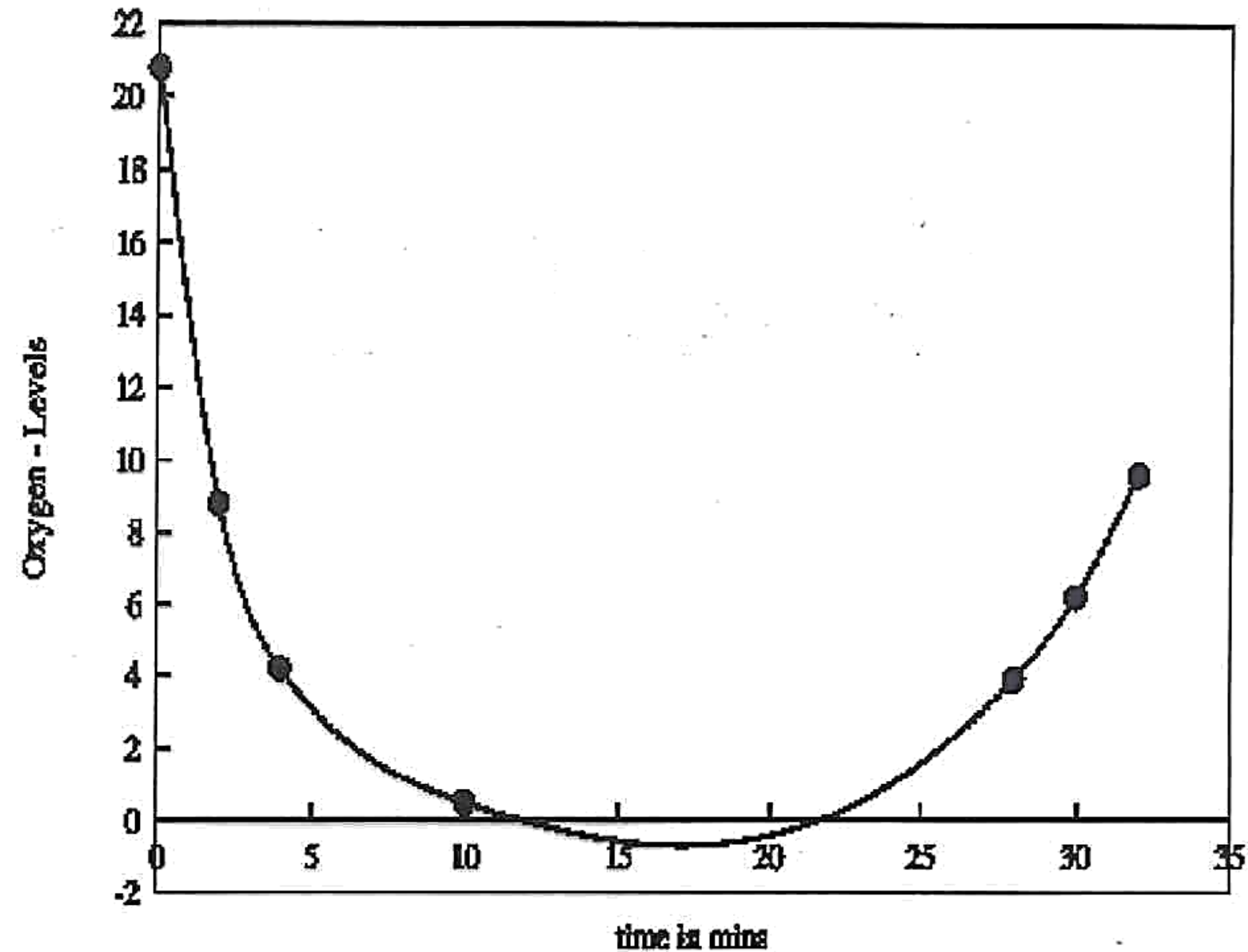
but is this what we want to see?

Visualizing the Data – Line Plot



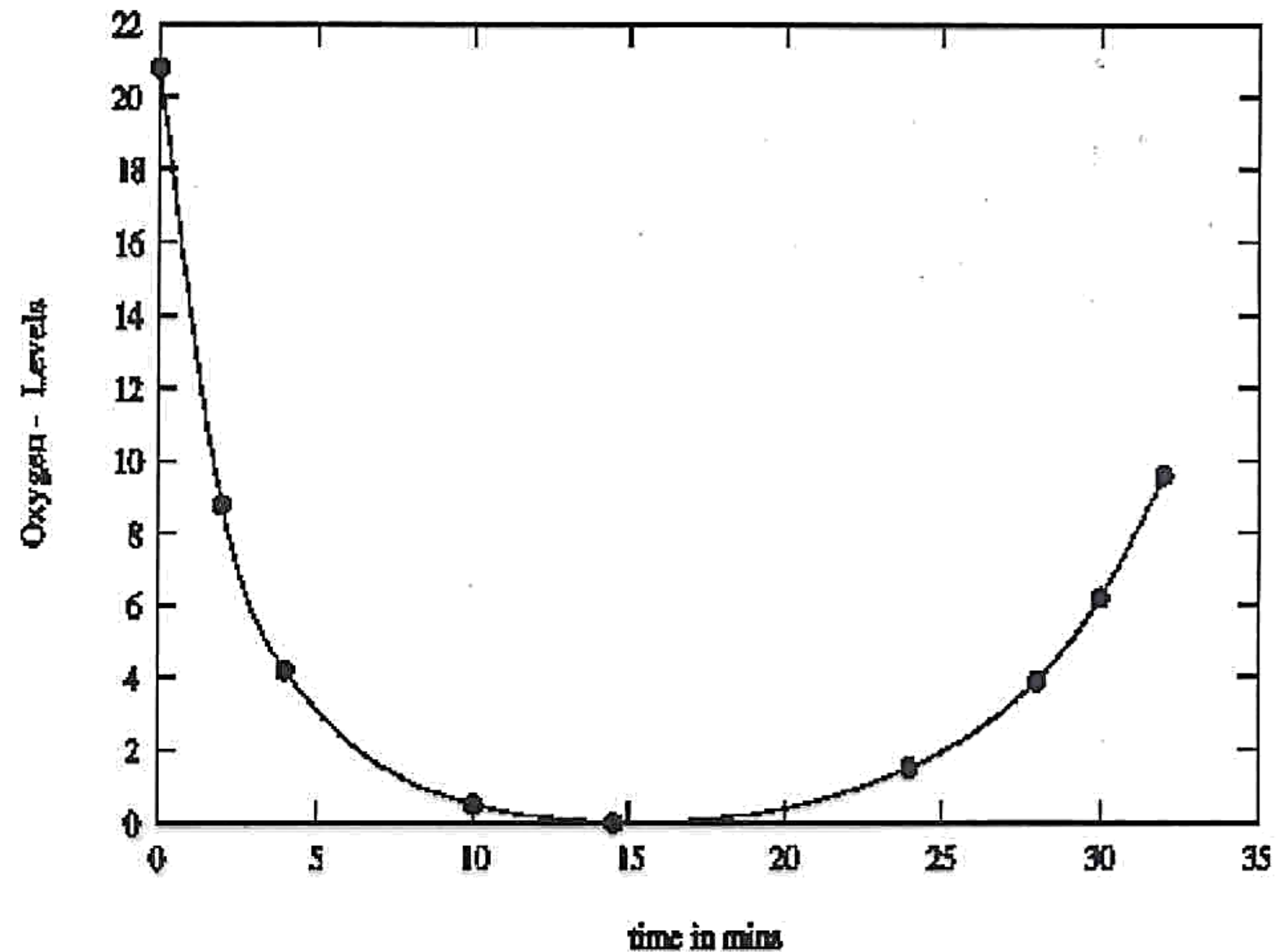
Estimating behaviour between the data - but is this believable?

Visualizing the Data – Curve Plot



Now it looks believable... but something is wrong

Visualizing the Data – Constrained Curve Plot



At least this is credible..

What Have We Learnt?

- It is not only the data that we wish to visualize - it is also the bits in between!
- The data are samples from some underlying 'field' which we wish to understand
- First step is to create from the data a 'best' estimate of the underlying field - we shall call this a **MODEL**
- This needs to be done with care and may need guidance from the scientist
- The process of fitting a continuous curve (or surface, or volume) through given data is known as **INTERPOLATION**

- Genealogical Graphs
- MatrixExplorer
- Flow Maps
- LiveRAC
- LinLog
- Sizing the Horizon

- The four nested levels of vis design that also facilitate for addressing validation at each level
- Specific design issues to consider at each level, their dependency and iterative nature of design choices between levels
- Specific validation threats at each level and the validation approaches should be chosen accordingly
- The nested model explicitly separates the vis design problem into levels in order to guide validation according to the unique threats at each level – to avoid mismatch